MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4702: MODELING AND SIMULATION

END OF SEMESTER EXAMINATION

DATE: SEPTEMBER, 2021 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

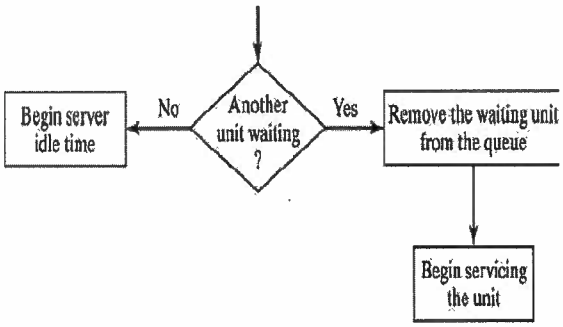
1. There are **5** questions in this paper. Answer **ALL**.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer..
5. This paper contains **FOUR** pages. Please check.

1. In the modeling and simulation of queueing systems

   a. What is the difference between queue behaviour and queue discipline? **[4 marks]**

   b. **Figure 1** shows two protocols (event flow diagrams) associated with a *G/G/1* queueing system.

Protocol A



Protocol B

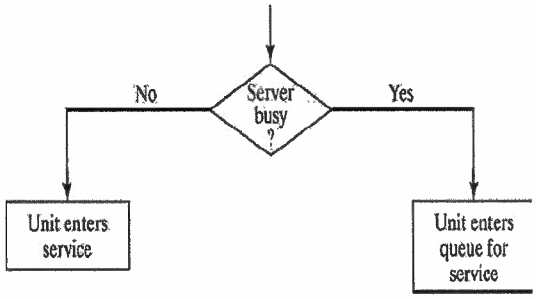


Figure 1

   i. Identify the two protocols **[2 marks]**

   ii. Interpret the two protocols, making necessary assumptions. **[4 marks]**

   iii. Develop related protocols for a *G/G/c* queueing system. **[10 marks]**

2. Students on **Chimangu University** campus arrive at a library counter at random times that are from 1 to 8 minutes apart. Each possible value of the interarrival time has the same probability of occurrence. The time for a transaction at the counter is distributed as follows.

Transaction time (Minutes)	1	2	3	4	5	6
Probability	0.10	0.20	0.30	0.25	0.10	0.05

The following *(studentIdentity, interArrivalTimeRandomDigit, transactionTimeRandomDigit)* triples represent the first 7 students. The list is sorted by the *studentIdentity* attribute.

(1, - , 84); (2, 064, 18); (3, 112, 87); (4, 678, 81); (5, 289, 06); (6, 871, 91); (7, 583, 79);

a. For the 7 students find their

i. Interarrival times. **[3 marks]**

ii. Service times. **[3 marks]**

b. Simulate the system for the 7 students, making necessary assumptions, by

i. Process interaction method. **[7 marks]**

ii. Student-specific event scheduling method assuming system termination override at 30 minutes. **[7 marks]**

3. Random numbers are a necessary basic ingredient in the simulation of almost all discrete systems.

a. Linear congruential method produces a sequence of integers, $X_1, X_2, \ldots$ by the following recursive relationship:

$$X_{i+1} = (aX_i + c) \bmod m, \quad i = 0, 1, 2, \ldots$$

Identify X_0, a, c and m **[4 marks]**

b. In order to improve on some of the desired characteristics of the generated random sequence, a combined linear congruential generator is defined as follows, where variables carry their usual meanings:

$$X_i = \left(\sum_{j=1}^{k} (-1)^{j-1} X_{i,j} \right) \bmod (m_1 - 1)$$

For 16-bit computers, L'Ecuyer recommends combining three multiplicative generators, with $m_1 = 32363$, $a_1 = 157$, $m_2 = 31727$, $a_2 = 146$, $m_3 = 31657$ and $a_3 = 142$. The period of the combined generator is approximately 8×10^{12}. Generate 5 random numbers with the combined generator, using the initial seeds $X_{i,0} = 100,\ 300,\ 500$, for the individual generators $i = 1, 2, 3$. **[8 marks]**

c. The random numbers 0.594, 0.928, 0.515, 0.055, 0.507, 0.351, 0.262, 0.797, 0.788, 0.442, 0.097, 0.798, 0.227, 0.127, 0.474, 0.825, 0.007, 0.182, 0.929 and 0.852 have been generated by an undocumented **DrgZ.m** routine of a simulation package. Use

Kolmogorov-Smirnov test with $\alpha = 0.05$ and a critical value $D_{0.05} = 0.294$ to learn whether the hypothesis that the numbers are uniformly distributed on the interval [0,1] can be rejected **[8 marks]**

4. Inverse Transform Technique is an algorithm for generating random variates for a desired probability distribution X given a sequence of random numbers R. The algorithm actually develops a random-variate generator $F^{-1}(X)$ as follows

---

Algorithm: Inverse Transform Technique

Compute cumulative distribution function (cdf) of desired variable X

Set $F(X) = R$ on the range X

Solve the equation $F(X) = R$ for X in terms of R

---

a. Using the Algorithm to determine random-variate generator for distributions with the following pdfs or cdfs:

i. $$f(x) = \begin{cases} e^{2x}, & -\infty < x \le 0 \\ e^{-2x}, & 0 < x \le \infty \end{cases}$$ **[4 marks]**

ii. $$F(X) = \begin{cases} 0, & x \le 0 \\ \frac{x^2}{2}, & 0 < x \le 1 \\ 1 - \frac{(2-x)^2}{2}, & 1 < x \le 2 \\ 1, & x > 2 \end{cases}$$ **[4 marks]**

b. On an opening day, a total of 100 students of Mzuzu University register for BICT 4702 Course as follows: 31 students register between 8 AM and 9 AM, 10 students register between 9 AM and 10 AM, 25 students register between 10 AM and 11 AM and 34 students register between 11 AM and 12 PM. Students' registration order is by their secretly assigned random numbers. At what time does a student with a random number $R = 0.830$ register for the course? **[10 marks]**

c. Describe Acceptance-Rejection Technique algorithm for generating random variates. **[3 marks]**

d. Guests arrive to a birthday party according to a Poisson process with a mean of a guest per 15 minutes. Through generation of a random variates, find the number of guests arriving during a 1-hourr time slot, assuming the following order of random numbers: 0.4357, 0.4146, 0.8353, 0.9952, 0.8004, 0.7945, 0.1530, 0.7454, 0.8291. **[7 marks]**

5. The quality of input to a system has effects on the performance of the system. Input modeling is a fundamental aspect to system simulation.

a. State **four** steps involved in input modeling **[4 marks]**

b. At Chimangu Hospital records pertaining to the monthly number of injuries were studied by a BICT 4702 student. The values for the past 100 months were as follows:

Injuries per Month	*Frequency of Occurrence*
0	35
1	40
2	13
3	6
4	4
5	1
6	1

Apply the chi-square test to these data to test the hypothesis that the underlying distribution is Poisson. Use $\chi^2_{ref} = 5.99$. **[8 marks]**

END OF QUESTION PAPER